1.3 SPORTS CHANGING FACILITY

LOCATION : West Coventry Academy GIA : 342 m²

Constructed at West Coventry Academy the Sports Changing Suite supports their existing levels of outdoor sports facilities. By locating this alternative changing accommodations at the heard of the outdoor sports areas it will be able to increase the offer of grass field based sports and the school can offer pupils the opportunity of created a community sports hub with the additional staff room and covered outdoor teaching space.

The suite is delivered using volumetric construction, incorporating a hybrid timber and steel frame. An Energy Pod is also included as part of the project, demonstrating how school IT and plumbing services can be upgraded quickly and efficiently using prefabricated volumetric solutions. T

OPERATIONAL CARBON

Operational carbon refers to the carbon emissions produced by a building or infrastructure during its use and operation. This includes emissions from energy used for heating, cooling, ventilation, lighting, and other operational needs.

Energy Use Intensity
67.0 kWh/yr/m²

Number PVs
88

PVs Annual Generation
118 kWh/yr/m²

PVs Contribution
8 l/s/p

EMBODIED CARBON

Embodied or embedded carbon is the carbon dioxide emitted in producing materials and products. It is a total of the energy used to extract and transport raw materials in addition to emissions associated with the manufacturing process. In construction embodied carbon is measured over the whole lifespan of a building, typically assumed to be 60 years. The carbon impact of the building's construction is calculated at each phase of the development chronologically. A1-A5 Upfront, production and construction, B1-B6 in use, and C1-C4 End of life. The carbon emissions associated with each stage are reported as kgCO2e/m². Biogenic carbon is the carbon stored within the timber elements. recycling or disposal as waste. The carbon emissions associated with each stage are reported as kgCO2e/m².

Substructure
73

Structure
96

Envelope
80

Internal fit out
64

MEP
132

A1-A5 Carbon